

REGULARISATION IN GUIDING

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1. SETUP

We consider the Markov kernel $P(x_0, dx_1)$ defined by

$$X_1 \mid X_0 \sim \mathcal{N}(\omega + \psi X_0, \eta^2).$$

We assume $v \sim x + \log Z^2$, where $Z \sim \mathcal{N}(0, 1)$. Define

$$h(x_1) = \text{density of } X_1 + \log Z^2 \text{ evaluated at } v, \quad \text{where } Z \sim \mathcal{N}(0, 1).$$

We have

$$P(X_1 + \log Z^2 \leq v) = P(Z^2 \leq e^{v-X_1})$$

so

$$h(x_1) = f_{\chi^2(1)}(e^{v-x_1}) e^{v-x_1}.$$

For large x_1 it will behave like $\sqrt{e^{v-x_1}}$. It should also behave this way for large x_1 . Define, using Gaussian approximation to the distribution of $\log Z^2$

$$g(x_1) = \varphi(v; \mu + x_1, \sigma^2), \quad \mu = -1.27, \quad \sigma^2 = \pi^2/2,$$

where $\varphi(\cdot; \mu, \sigma^2)$ denotes the Gaussian density.

We have twisted kernels

$$P^h(x_0, dx_1) = \frac{P(x_0, dx_1) h(x_1)}{\int P(x_0, dx_1) h(x_1)}, \quad P^g(x_0, dx_1) = \frac{P(x_0, dx_1) g(x_1)}{\int P(x_0, dx_1) g(x_1)}.$$

Define $c_g = \int P(x_0, dx_1) g(x_1)$ and similarly c_h . In our specific case

$$c_g = \varphi(v; \mu + \omega + \psi x_0; \sigma^2 + \eta^2).$$

Note

$$P^{g+\varepsilon}(x_0, dx_1) = \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = \lambda_g(\varepsilon) P^g(x_0, dx_1) + (1 - \lambda_g(\varepsilon)) P(x_0, dx_1),$$

with $\lambda_g(\varepsilon) = c_g/(c_g + \varepsilon)$.

2. SEQUENTIAL MONTE CARLO APPROACH

Suppose we do sequential Monte Carlo to sample from $X_1 \mid v$ and to find the marginal likelihood $p(v)$, which is actually c_h . We use mutation kernel $M = P^{g+\varepsilon}$. Then it follows from the results in Chapter 5 of Chopin-Papaspiliopoulos that the potential should be

$$\begin{aligned} G_\varepsilon(x_1) &:= h(x_1) \frac{P(x_0, dx_1)}{P^{g+\varepsilon}(x_0, dx_1)} = h(x_1) \frac{\int P(x_0, dx_1) g(x_1) dx_1 + \varepsilon}{g(x_1) + \varepsilon} \\ &= h(x_1) \frac{c_g + \varepsilon}{g(x_1) + \varepsilon}. \end{aligned}$$

The estimator for the marginal likelihood is

$$\hat{L} := \frac{1}{N} \sum_{n=1}^N G_\varepsilon(X_1^{(n)}), \quad (X_1^{(n)})_{n=1}^N \sim P^{g+\varepsilon}(x_0, dx_1).$$

For *any* $\varepsilon > 0$

$$\mathbb{E}_{P^{g+\varepsilon}} \hat{L} = \mathbb{E}_{P^{g+\varepsilon}} G_\varepsilon(X_1) = \int h(x_1) \frac{c_g + \varepsilon}{g(x_1) + \varepsilon} \frac{g(x_1) + \varepsilon}{c_g + \varepsilon} P(x_0, dx_1) = c_h.$$

2.1. Analysis of variance of $\hat{L}$. Define

$$m(\varepsilon) = \mathbb{E}_{P^{g+\varepsilon}} [G_\varepsilon(x_0, X_1)]^2.$$

By expanding we get

$$\begin{aligned} m(\varepsilon) &= \int h(x_1)^2 \left(\frac{c_g + \varepsilon}{g(x_1) + \varepsilon} \right)^2 \frac{P(x_0, dx_1) (g(x_1) + \varepsilon)}{c_g + \varepsilon} \\ &= (c_g + \varepsilon) \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1). \end{aligned}$$

Define

$$I(\varepsilon) = \int \frac{h(x_1)^2}{g(x_1) + \varepsilon} P(x_0, dx_1)$$

and

$$\bar{m}(\varepsilon) = \log m(\varepsilon) = \log(c_g + \varepsilon) + \log I(\varepsilon).$$

Note that we can rewrite to

$$I(\varepsilon) = \int h(x_1) \frac{\int h(x_1) P(x_0, dx_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1) = c_h \int \frac{h(x_1)}{g(x_1) + \varepsilon} P^h(x_0, dx_1).$$

though we won't use this expression in the numerics.

2.2. Infinite variance case. Suppose $\varepsilon = 0$. Since $c > 0$, finiteness of $\bar{m}(0)$ is equivalent to finiteness of

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1).$$

Using $f_{\chi^2(1)}(u) = (2\pi u)^{-1/2} e^{-u/2}$ we get

$$h(x_1) = \frac{1}{\sqrt{2\pi}} e^{-(v-x_1)/2} e^{-e^{v-x_1}/2},$$

so

$$h(x_1)^2 = \frac{1}{2\pi} e^{-(v-x_1)} e^{-e^{v-x_1}}.$$

Writing $t = x_1 - v$, the integrand of $I(0)$ is proportional to

$$\frac{h(x_1)^2}{g(x_1)} p(x_1 | x_0) \propto \exp\left(t - e^{-t} + \frac{(t + \mu)^2}{2\sigma^2} - \frac{(t - (\omega + \psi x_0 - v))^2}{2\eta^2}\right).$$

We analyse the tails:

- *Right tail* $t \rightarrow +\infty$. The term $e^{-t} \rightarrow 0$, and expanding the quadratics the dominant behaviour is

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right).$$

This is integrable if and only if the coefficient of t^2 is strictly negative, i.e.

$$\frac{1}{2\sigma^2} - \frac{1}{2\eta^2} < 0 \iff \eta^2 < \sigma^2.$$

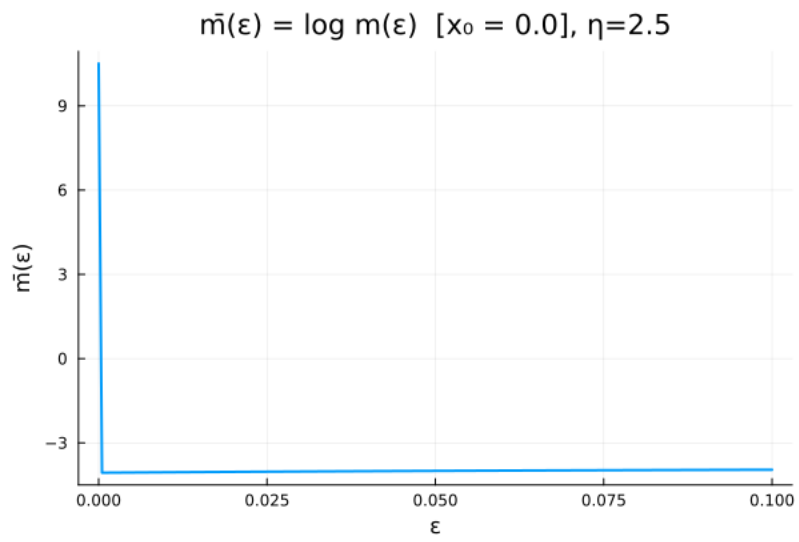
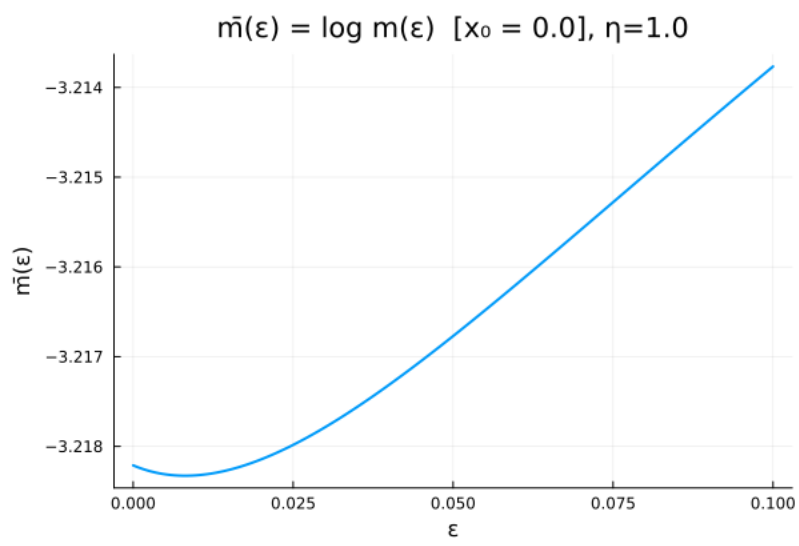
- *Left tail* $t \rightarrow -\infty$. The term $e^{-t} \rightarrow +\infty$ dominates inside the exponent, sending the integrand to zero faster than any polynomial. Hence the left tail never causes divergence.

Therefore,

Proposition 1. With the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0, 1)$, we have

$$I(0) = \int \frac{h(x_1)^2}{g(x_1)} P(x_0, dx_1) < \infty \iff \eta < \sigma.$$

It follows that for $\eta < \pi/\sqrt{2} \approx 2.221$ no regularisation is needed. For $\eta = \sigma$ the integral diverges (though like e^t), while for $\eta > \sigma$ divergence is much faster.



3. IMPORTANCE SAMPLING QUALITY: KL AND χ^2 DIVERGENCES

3.1. General theory. The natural divergences for assessing IS quality of $\hat{L}$ are the KL divergence and the χ^2 divergence between the target P^h and the proposal $P^{g+\varepsilon}$:

$$\begin{aligned}\text{KL}(P^h \| P^{g+\varepsilon}) &= \int \log \frac{dP^h}{dP^{g+\varepsilon}} dP^h, \\ \chi^2(P^h \| P^{g+\varepsilon}) &= \int \left(\frac{dP^h}{dP^{g+\varepsilon}} \right)^2 dP^{g+\varepsilon} - 1.\end{aligned}$$

Since

$$\frac{dP^h}{dP^{g+\varepsilon}}(x_1) = \frac{h(x_1)/c_h}{(g(x_1) + \varepsilon)/(c_g + \varepsilon)} = \frac{G_\varepsilon(x_1)}{c_h},$$

we have

$$\chi^2(P^h \| P^{g+\varepsilon}) = \frac{m(\varepsilon)}{c_h^2} - 1,$$

so finite variance of the IS estimator is *exactly equivalent* to $\chi^2(P^h \| P^{g+\varepsilon}) < \infty$. The χ^2 divergence thus directly controls the variance, while the KL divergence controls the asymptotic quality of log-likelihood estimators (see Diaconis et al.).

3.2. KL divergence: explicit form. We have

$$\text{KL}(P^h \| P^{g+\varepsilon}) = \log \frac{c_g + \varepsilon}{c_h} + \int \log h(x_1) P^h(x_0, dx_1) - \int \log(g(x_1) + \varepsilon) P^h(x_0, dx_1).$$

As $\varepsilon \downarrow 0$ each term has a finite limit provided $\int |\log g(x_1)| P^h(x_0, dx_1) < \infty$. Since $\log g(x_1)$ is quadratic in x_1 (as g is Gaussian), this requires P^h to have finite second moments. Hence

$$\text{KL}(P^h \| P^{g+\varepsilon}) \rightarrow \text{KL}(P^h \| P^g) < \infty \quad \text{as } \varepsilon \downarrow 0,$$

whenever P^h has finite second moments, which holds for any $\varepsilon \geq 0$ in our specific example.

3.3. χ^2 divergence and finite variance: the threshold. For any $\varepsilon > 0$, the denominator $g(x_1) + \varepsilon \geq \varepsilon > 0$ uniformly, so

$$m(\varepsilon) \leq \varepsilon^{-1} \int h(x_1)^2 P(x_0, dx_1) < \infty,$$

provided $h \in L^2(P(x_0, \cdot))$, which holds for our specific h . Hence the IS estimator has finite variance for *every* $\varepsilon > 0$.

At $\varepsilon = 0$ the situation is more delicate. From the tail analysis of Section 2, the integrand $h(x_1)^2/g(x_1)$ behaves like

$$\exp\left(t^2\left(\frac{1}{2\sigma^2} - \frac{1}{2\eta^2}\right) + O(t)\right) \quad \text{as } t = x_1 - v \rightarrow +\infty,$$

leading to the following result.

Proposition 2. For the observation scheme $v = X_1 + \log Z^2$, $Z \sim \mathcal{N}(0, 1)$:

$$\chi^2(P^h \| P^g) < \infty \iff m(0) < \infty \iff \eta < \sigma.$$

Now

$$ESS \approx \frac{N}{1 + \chi^2(P^h \| P^{g+\varepsilon})}.$$

So minimising $m(\varepsilon)$ at each SMC step is equivalent to maximising ESS.

4. NUMERICAL EXPERIMENT: EFFECT OF ε ON SMC PERFORMANCE

4.1. **Setup.** We consider the hidden Markov model

$$\begin{aligned} X_t \mid X_{t-1} &\sim \mathcal{N}(\omega + \psi X_{t-1}, \eta^2), \\ V_t \mid X_t &= X_t + \log Z_t^2, \quad Z_t \sim \mathcal{N}(0, 1), \end{aligned}$$

with parameters $\omega = 0$, $\psi = 0.9$, $\mu = -1.27$, $\sigma^2 = \pi^2/2$. We compare two regimes:

- **Regime A** ($\eta = 1.0 < \sigma \approx 2.22$): finite-variance regime, where $m(0) < \infty$.
- **Regime B** ($\eta = 3.0 > \sigma$): infinite-variance regime, where $m(0) = \infty$.

In both regimes we run SMC with the twisted proposal $P^{g+\varepsilon}$ for several values of $\varepsilon \in \{0, 0.001, 0.01, 0.05, 0.1\}$.

4.2. **SMC Algorithm.** At each time step t , the twisted proposal is the mixture

$$P^{g+\varepsilon}(x, \cdot) = \frac{c(x)}{c(x) + \varepsilon} P^g(x, \cdot) + \frac{\varepsilon}{c(x) + \varepsilon} P(x, \cdot),$$

where both $c(x)$ and the parameters of P^g are available in closed form (see Section 2). The IS weight for a particle x_t proposed from $P^{g+\varepsilon}(x_{t-1}, \cdot)$ is

$$G_\varepsilon(x_{t-1}, x_t) = \frac{h_t(x_t)}{g_t(x_t) + \varepsilon} (c(x_{t-1}) + \varepsilon).$$

Algorithm 1 SMC with twisted proposal $P^{g+\varepsilon}$

Require: observations $v_{1:T}$, particle count N , perturbation $\varepsilon \geq 0$

- 1: Initialise particles $x_0^{(i)} \sim \mathcal{N}(0, \eta^2/(1 - \psi^2))$ for $i = 1, \dots, N$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: **Propose:** for each i , draw

$$x_t^{(i)} \sim P^{g_t+\varepsilon}(x_{t-1}^{(i)}, \cdot)$$

by sampling from $P^{g_t}(x_{t-1}^{(i)}, \cdot) = \mathcal{N}(\mu^*, \sigma^{*2})$ with probability $\alpha^{(i)} = c(x_{t-1}^{(i)})/(c(x_{t-1}^{(i)}) + \varepsilon)$, and from $P(x_{t-1}^{(i)}, \cdot)$ otherwise

- 4: **Weight:** compute unnormalised weights

$$w^{(i)} = \frac{h_t(x_t^{(i)})}{g_t(x_t^{(i)}) + \varepsilon}$$

- 5: **Likelihood increment:**

$$\log \hat{p}(v_t \mid v_{1:t-1}) = \log(\bar{c}_t + \varepsilon) + \log \frac{1}{N} \sum_{i=1}^N w^{(i)}, \quad \bar{c}_t = \frac{1}{N} \sum_{i=1}^N c(x_{t-1}^{(i)})$$

- 6: **ESS:** compute $\text{ESS}_t = (\sum_i \tilde{w}^{(i)2})^{-1}$ where $\tilde{w}^{(i)} = w^{(i)} / \sum_j w^{(j)}$
 - 7: **if** $\text{ESS}_t < N/2$ **then**
 - 8: **Resample** particles using systematic resampling with weights $\tilde{w}^{(i)}$
 - 9: **end if**
 - 10: **end for**
 - 11: **return** $\sum_{t=1}^T \log \hat{p}(v_t \mid v_{1:t-1})$, $\{\text{ESS}_t\}_{t=1}^T$
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To assess performance, we run Algorithm 1 independently $R = 200$ times on the same simulated dataset ($T = 50$, $N = 2000$) and record the minimum ESS over time,

$$\text{ESS}_{\min} = \min_{t=1,\dots,T} \text{ESS}_t,$$

for each run. A low $\text{ESS}_{\min}$ indicates that at some point the particle cloud collapsed to a near-degenerate distribution, with a single particle carrying almost all the weight.

4.3. Results. Figure 1 shows the empirical CDF of $\text{ESS}_{\min}$ across $R = 200$ runs for each value of ε , separately for the two regimes.

Regime A ($\eta < \sigma$). All curves are tightly clustered regardless of ε , with $\text{ESS}_{\min}$ concentrated between 400 and 600. This confirms the theoretical prediction: when $m(0) < \infty$, the IS weights have finite variance and ε has no meaningful effect on performance. Note though, while finiteness is guaranteed, it is typically beneficial to take nonzero ε . The value $m(0)$ can still be large, while finite.

Regime B ($\eta > \sigma$). The picture changes dramatically. At $\varepsilon = 0$ the CDF has substantial mass near zero, meaning that in a significant fraction of runs the particle cloud collapses to near-degeneracy at some time step. A small perturbation $\varepsilon = 0.001$ shifts the CDF markedly rightward, essentially eliminating the collapses. This is the direct numerical manifestation of $m(0) = \infty$: the IS weights have infinite variance, leading to occasional but catastrophic weight concentration on a single particle.

Figure 2 shows the mean and 5th percentile of $\text{ESS}_{\min}$ as a function of ε for both regimes. In Regime B, the 5th percentile at $\varepsilon = 0$ is near 0 out of 2000, confirming that in the worst 5% of runs the particle cloud is essentially degenerate. Both diagnostics improve sharply as ε increases from 0, reach a maximum around $\varepsilon^* \approx 0.001$, and then slowly deteriorate for larger ε . This non-monotone behaviour reflects two competing effects: for $\varepsilon < \varepsilon^*$, increasing ε regularises the weights and reduces the chance of collapse; for $\varepsilon > \varepsilon^*$, the proposal $P^{g+\varepsilon}$ moves away from the optimal twist P^g toward the untwisted kernel P , which is a poor proposal for h .

4.4. On the difficulty of choosing ε in Regime B. The results highlight that the choice of ε in Regime B is delicate. The optimal ε^* exists and can in principle be found by minimising $m(\varepsilon)$ numerically, as described in Section 3. However, $m(\varepsilon)$ depends on x_0 and varies across time steps as the particle cloud evolves, so a single fixed ε cannot be globally optimal. Furthermore, the performance is sensitive to this choice: the 5th percentile of $\text{ESS}_{\min}$ drops sharply on both sides of ε^* , leaving a narrow window of well-performing values.

This suggests that a fixed ε is fundamentally limited in Regime B, and that an *adaptive* scheme — choosing ε_t at each step based on the current particle cloud by minimising an empirical estimate of $m(\varepsilon)$ — would be more robust. In Regime A, by contrast, $\varepsilon = 0$ is already optimal and no adaptation is needed.

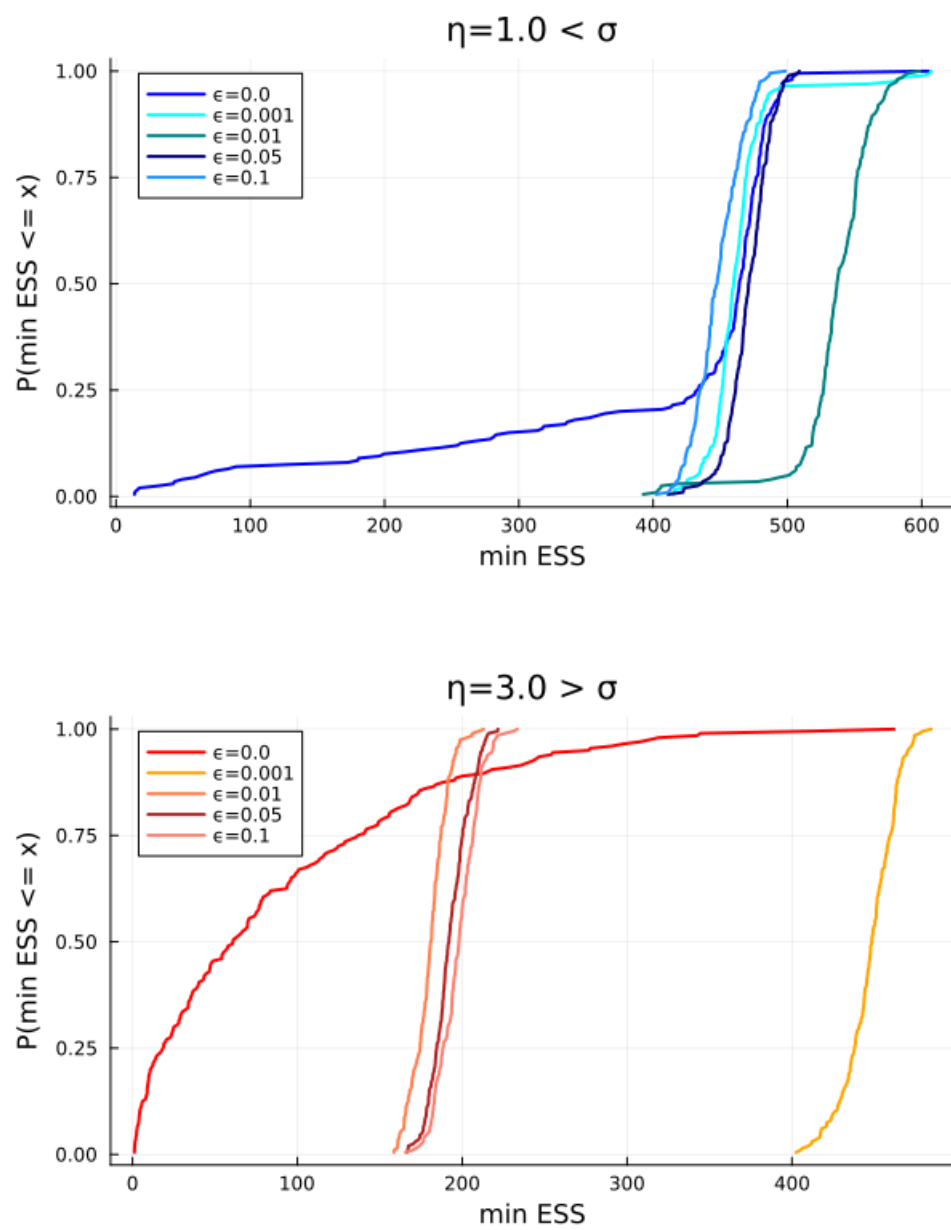


FIGURE 1.

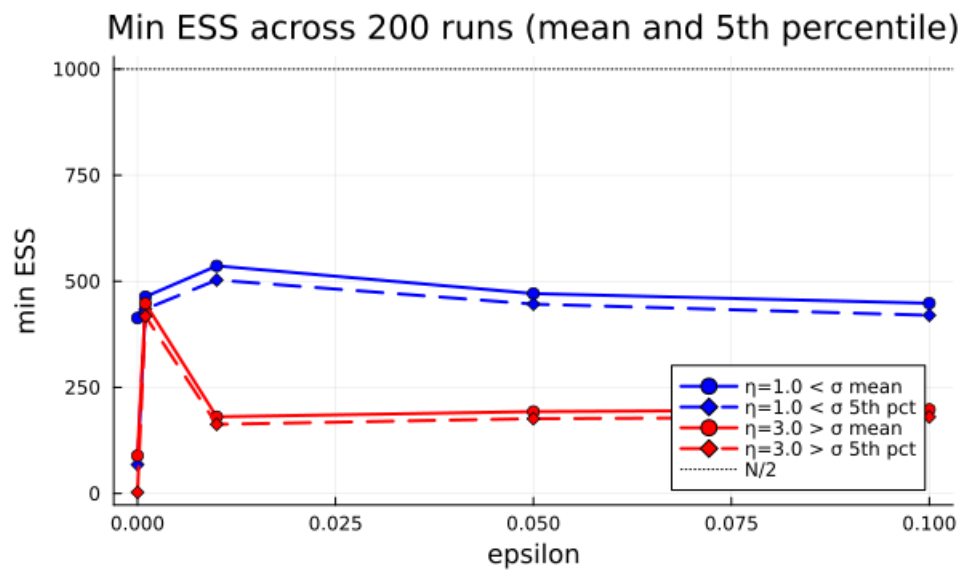


FIGURE 2.